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Problem Statement

Despite the rapid advancement of Generative AI in education, there is still no unified, multimodal, voice-enabled, and document-grounded AI learning ecosystem that adapts dynamically to individual student needs.

- Most AI tools are text-based chatbots.
- Limited integration of:
  - Multimodal learning (text +image +audio)
  - Voice tutoring systems
  - Progress tracking dashboards
  - Document-grounded responses (RAG)
- Students still manually:
  - Convert PDFs to notes
  - Create quizzes
  - Track learning progress

Therefore, there exists a gap between theoretical AI capabilities and practical personalized learning systems.

Motivation

•Students waste time converting PDFs into notes

•Lack of personalized learning systems

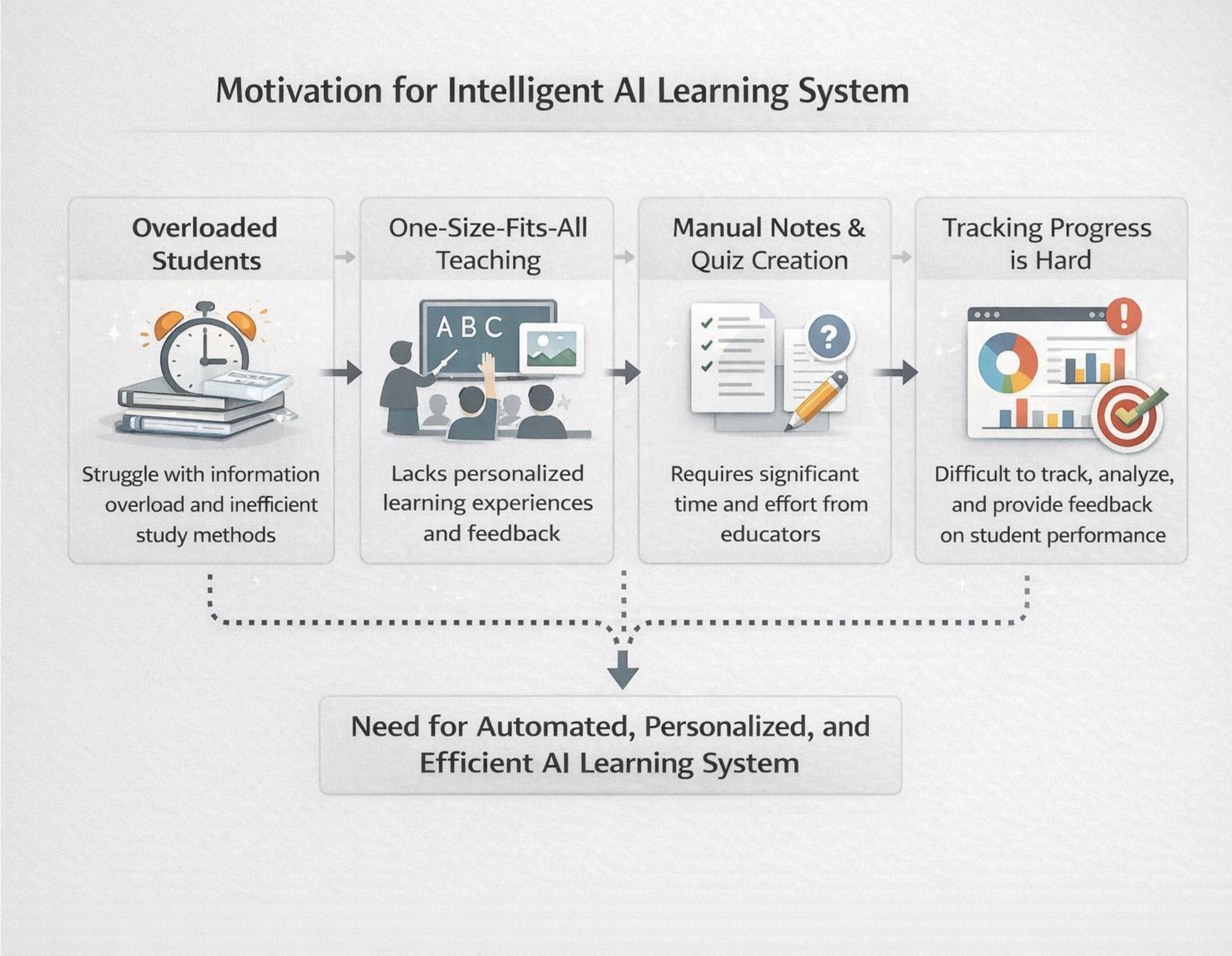
•Growing demand for **AI-based education tools**

•Need for:

Smart tutoring

Automated content generation

Adaptive learning systems



Introduction

Our Purpose is to build :

An AI-powered Study Companion that transforms raw learning material into an adaptive, multimodal and personalized which better the users learning experience.

What We Plan to Build:

A full-stack intelligent learning platform that:

1. Converts PDFs, text, URLs into:
   - Structured summaries
   - Flashcards
   - Interactive quizzes
2. Supports:
   - Visual learning (image generation)
   - Auditory learning (Text-to-Speech)
3. Tracks:
   - Learning progress
   - Weak topics
   - Study patterns
4. Uses:
   - RAG (Retrieval-Augmented Generation)
   - Prompt-engineered role-based AI
5. Provides:
   - Grounded, personalized, adaptive tutoring

Objectives

•Build a **multimodal AI system**

•Integrate:

RAG (document grounding)

Voice (Text-to-Speech)

Image Gen(Stable Diffusion)

Document Summerization

•Automate:

Notes generation

Quiz creation

•Track student progress

Literature Review

Paper	Focus Area	Methadology	Key Findings	Limitation
ChatISA	In-house AI chatbot for IS education	Design Science Research (DSRM), multi-model integration, Retrieval-Augmented Generation (RAG), open-source deployment	Domain-specific chatbot improves student interaction	Limited generalization, domain restriction
EDUBOT	AI-powered student assistant chatbot	GPT-4 integration, Google Custom Search API, performance visualization (Matplotlib), supervised evaluation	Improved engagement & academic performance	API dependency, limited multimodal depth
Generative AI in ITS	AI-enhanced Intelligent Tutoring Systems	Large Language Models (LLMs), automated question generation, adaptive feedback mechanisms	Dynamic question creation & personalized feedback	Pedagogical accuracy concerns, text-dominant interaction
LMFMs in Education	Large Multimodal Foundation Models	Multimodal transformer models, API integration, middleware architecture connecting AI systems	Rich multimodal interaction, enhanced engagement	Digital divide, hallucination risk, need for human oversight
Guettala et al., 2024	Generative AI in adaptive & personalized learning	Systematic literature review, case study analysis, framework-based evaluation of AI educational systems	Improved engagement, adaptive content, better learning outcomes	Ethical concerns, hallucination, lack of real-world deployment

Key Insights from Literature

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•Most systems are **chatbot-based**

•Limited **multimodal capabilities**

•Lack of **voice-enabled tutoring**

•No integrated **progress tracking systems**

•Heavy reliance on APIs

•Accuracy & hallucination issues exist

Limitations of Existing Systems

Domain-specific solutions (ChatISA)

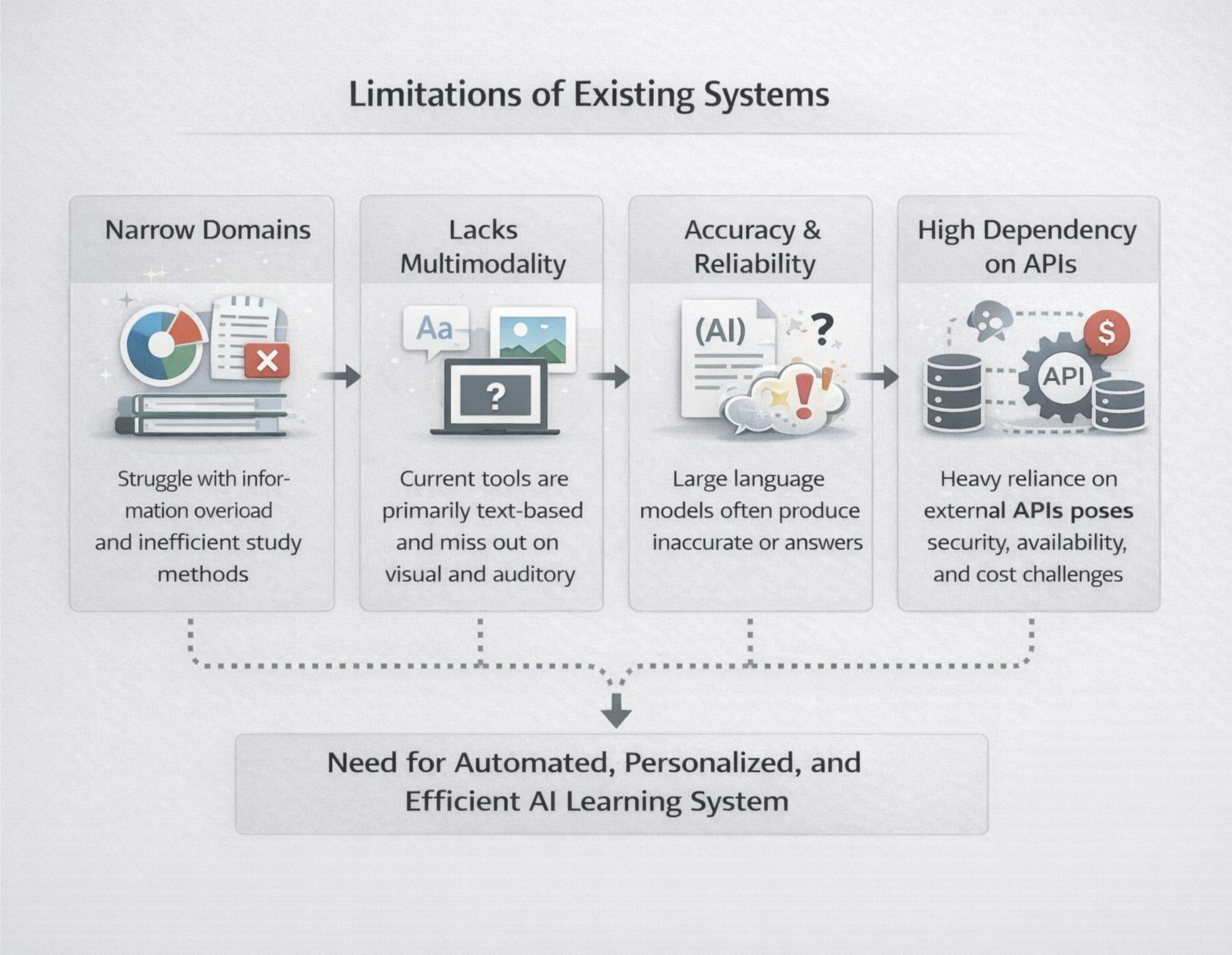
API dependency (EDUBOT)

Text-only systems (ITS)

Hallucination risks (LMFMs)

No real-world deployment (Guettala et al.)

👉 **No system integrates all features together**



Research Gap

- Absence of:
    - Multimodal + Voice + RAG integration
    - Personalized tracking system
    - Real-time adaptive learning
- Existing solutions are **fragmented**

Problem We Address

- Develop a system that:

  Combines **multimodal + voice + RAG**

  Provides **accurate, grounded responses**

  Tracks learning performance

- Goal: Build a **complete AI learning ecosystem**

Our work and Implementation

After recognizing the limitations of previous studies, we propose a novel solution to overcome these challenges : A Multimodal, Voice-Enabled, RAG-Grounded AI Study Companion with Full Progress Analytics

Key Modules:

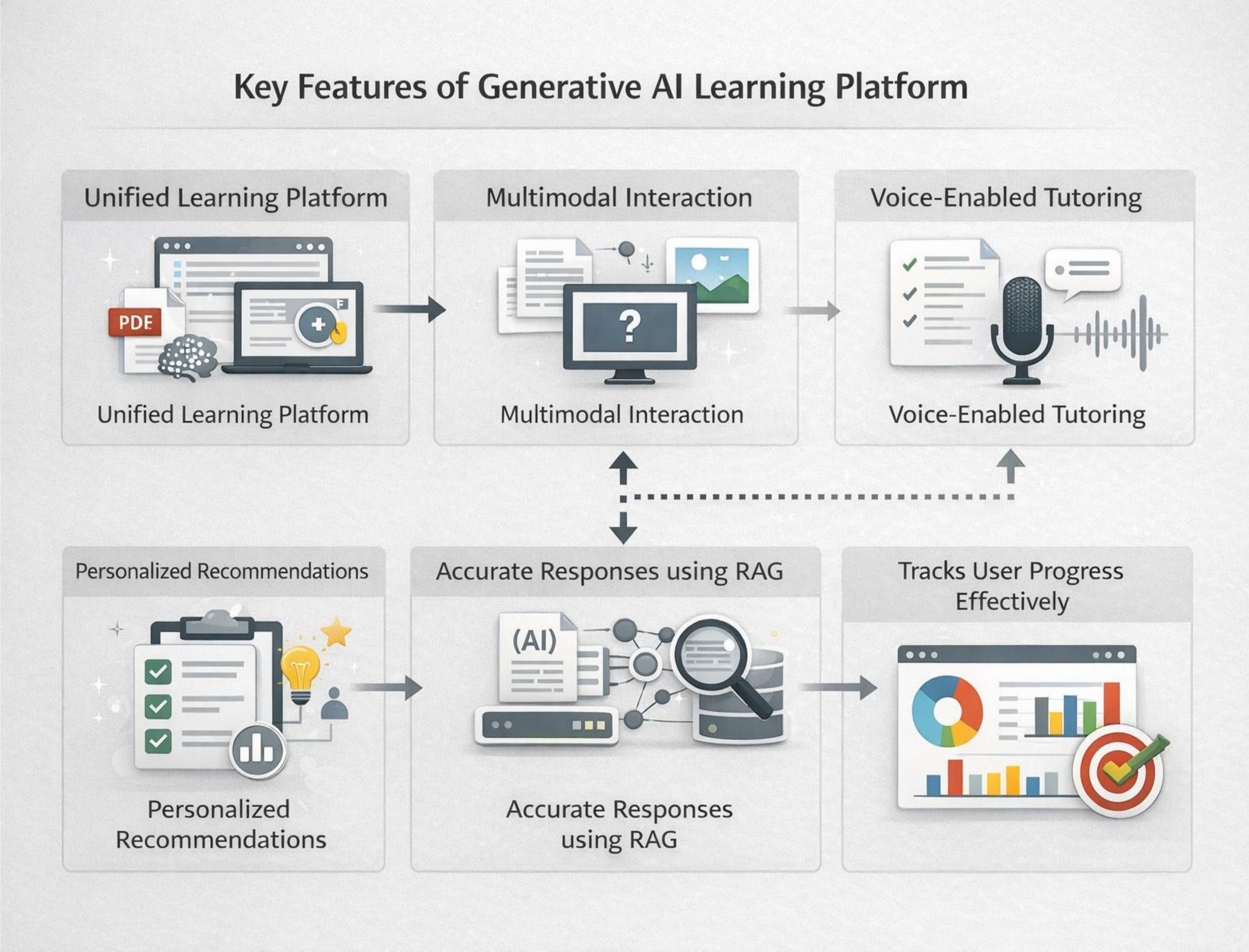
1. Data Ingestion
2. RAG Retrieval System
3. Prompt Engineering Layer
4. Multimodal Output Layer
5. Progress Tracking Engine
6. Backend-Frontend Integration

Research Gap We Address:

Identified Limitation	Our Solution
Text-only systems	Multimodal learning (Text + Image + Voice)
Hallucination	RAG-based document grounding
No voice tutoring	Integrated Text-to-Speech
No personalization tracking	Database-based analytics

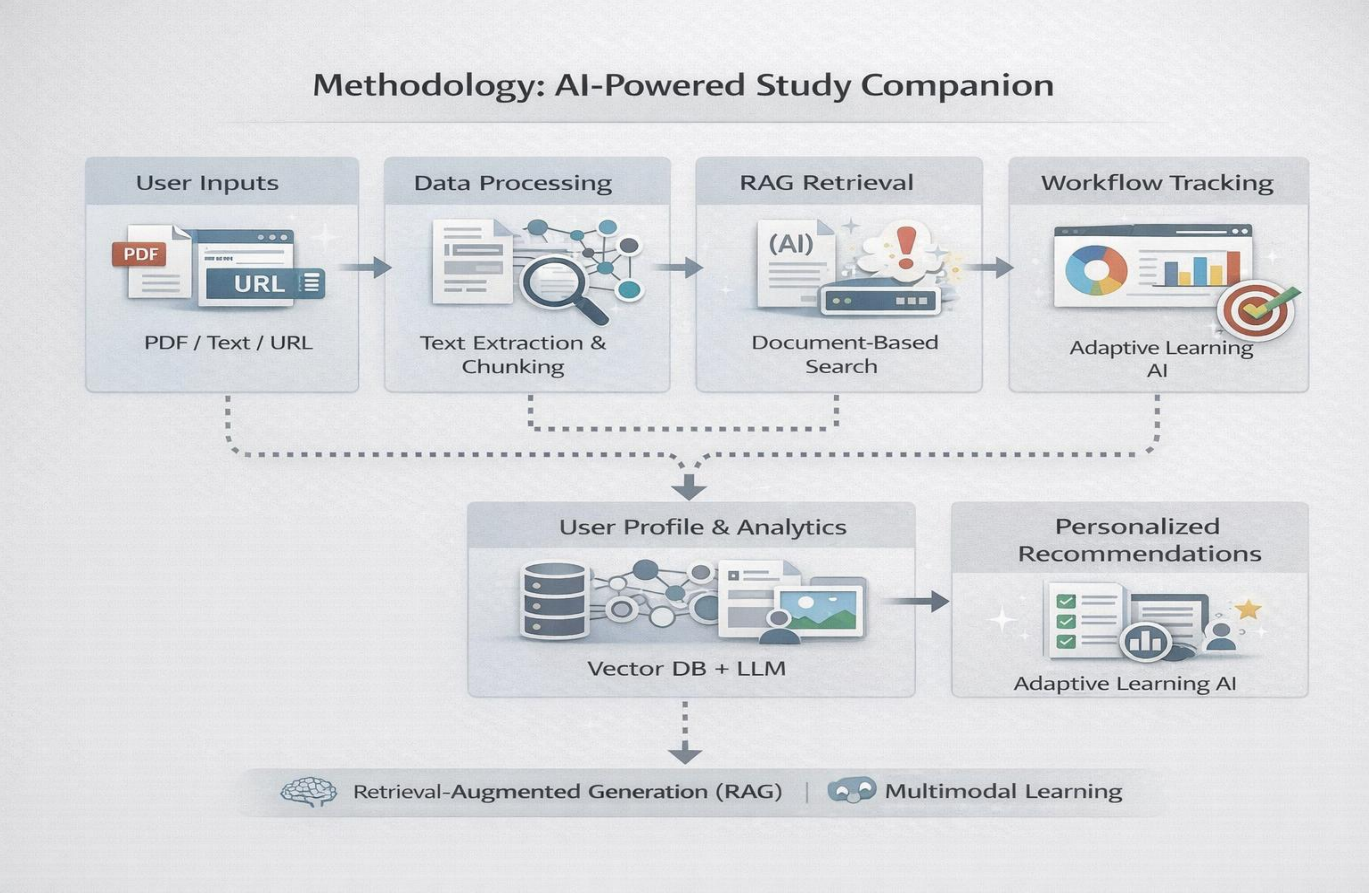
Key Features

- Multimodal learning (Text + Image + Audio)
- Voice-based tutoring
- RAG-based accurate responses
- Automated quizzes & summaries
- Progress tracking dashboard



Methodology

- User Input (PDF / URL / Topic)
- Data Processing
- RAG Retrieval
- Prompt Engineering
- Output Generation
- Progress Tracking



Workflow

Step 1: User Input
- Upload PDF / Enter URL / Enter Topic

Step 2: Data Processing
- Text extraction
- Chunking
- Embedding generation
- Store in vector database

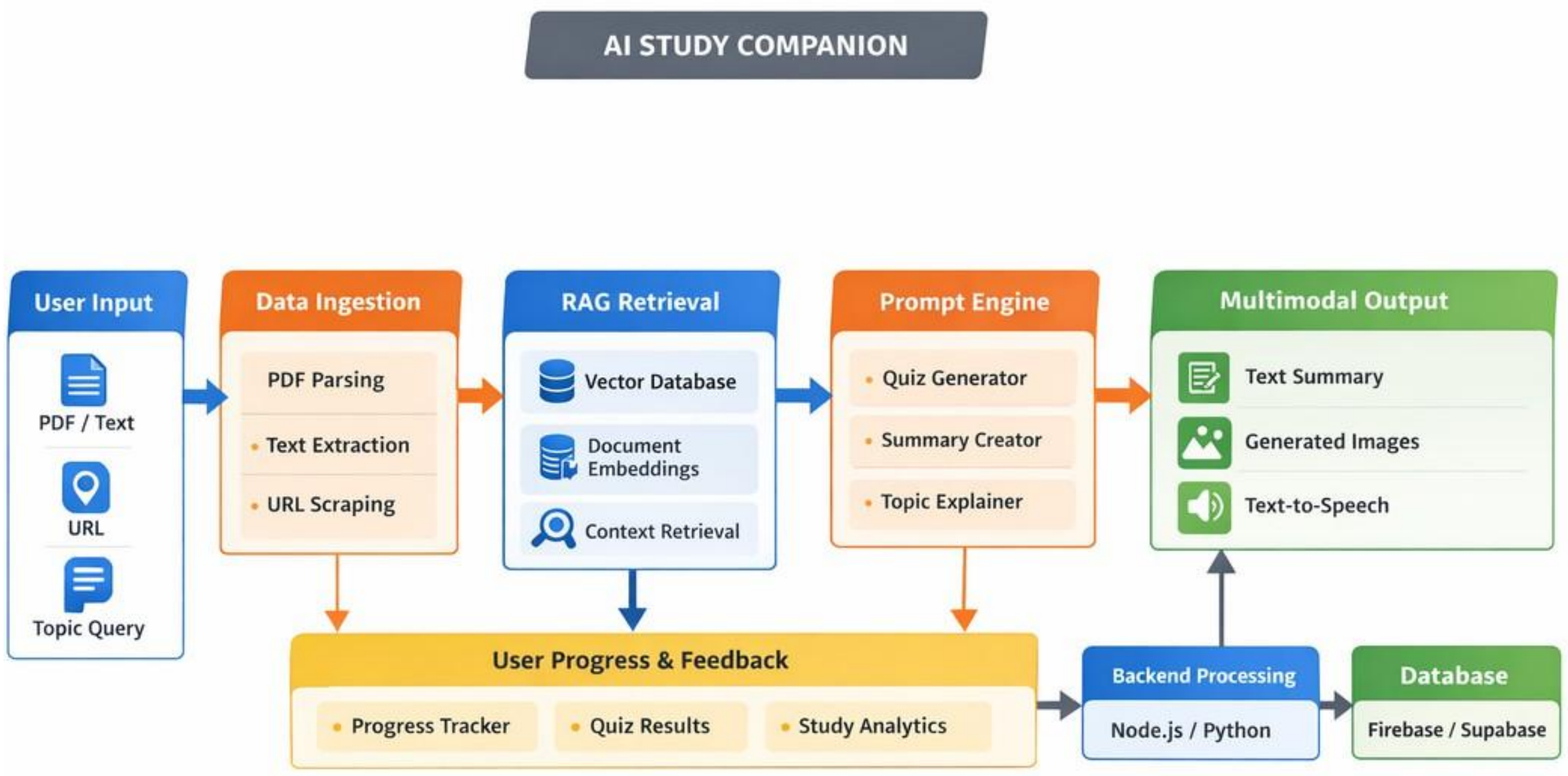
Step 3: RAG Retrieval
- Retrieve relevant document chunks
- Inject into prompt context

Step 4: Prompt Engine
- System role selection
- Generate:
  - Summary
  - Quiz
  - Explanation
  - Interview questions

Step 5: Multimodal Layer
- Generate diagram/image
- Convert explanation to voice
- Provide downloadable notes

Step 6: Progress Tracking
- Store:
  - Quiz scores
  - Study time
  - Weak topics
- Visual dashboard analytics

Step 7: Feedback Loop
- Use performance data
- Adjust difficulty
- Recommend revision topics



Base Paper Overview - Guettala et al., 2024

Title: Generative Artificial Intelligence in Education: Advancing Adaptive and Personalized Learning

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The paper explores:

- Evolution from AI →ML →DL →Generative AI
- Integration of Generative AI in adaptive learning
- environments Case studies of AI-driven educational systems
- Transition from Education 4.0 to Education 5.0

It focuses on:

- Personalized learning
- pathways Dynamic content generation Engagement improvement

Key Findings:

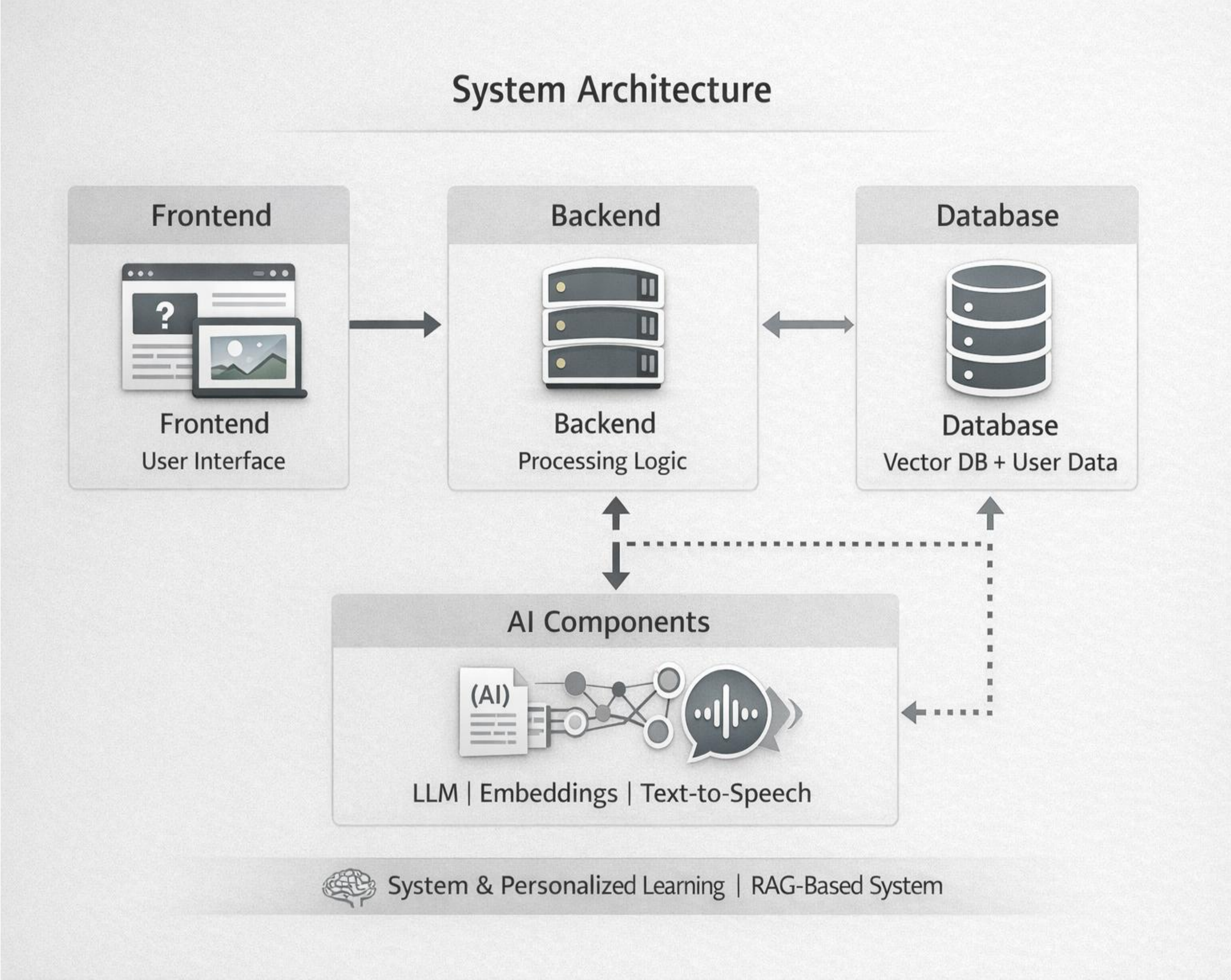
- Increased student
- engagement Improved test
- performance Accelerated
- skill development
- Personalized adaptive content improves retention

Limitations:

- Ethical & bias
- concerns
- Hallucination risks
- Lack of grounding mechanisms
- Limited multimodal real deployment

System Architecture

- Frontend: User Interface
- Backend: Processing engine
- Database:
  - Vector DB (for RAG)
  - User data storage
- AI Models:
  - LLM
  - Embeddings
  - Text-to-Speech



Modules of System

Key Modules:

1. Data Ingestion
2. RAG Retrieval System
3. Prompt Engineering Layer
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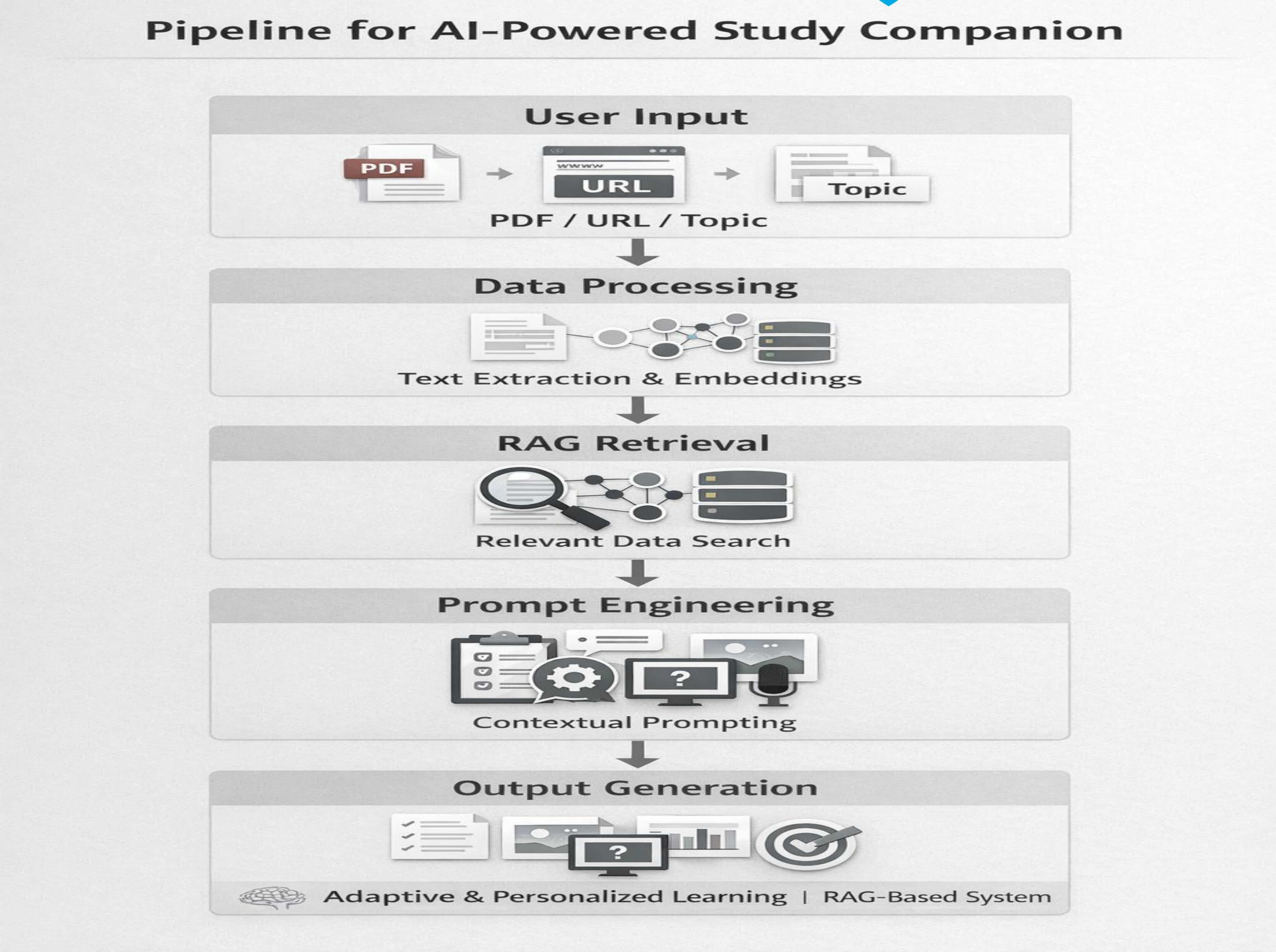


Technologies Used

Authentation Pass key for login

Database used for storing vector chunks

RAG used for personalized querries

Quiz generator by using open router api

Image generation using Stable Diffusion

Text to Voice using gTTs library

Code Explainer and generator

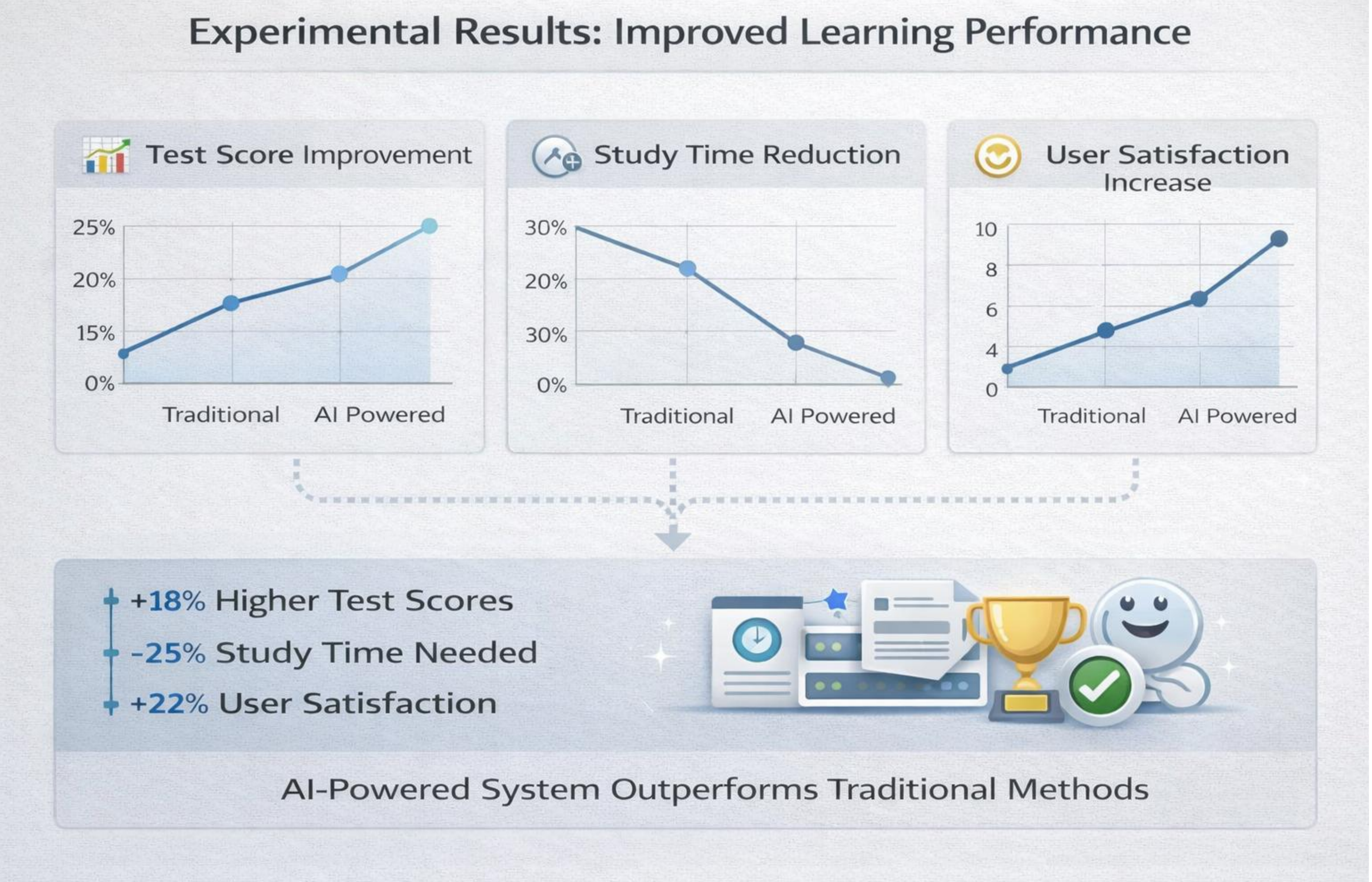
Rag based voice to voice

Pdf to Word and vice versa conversion

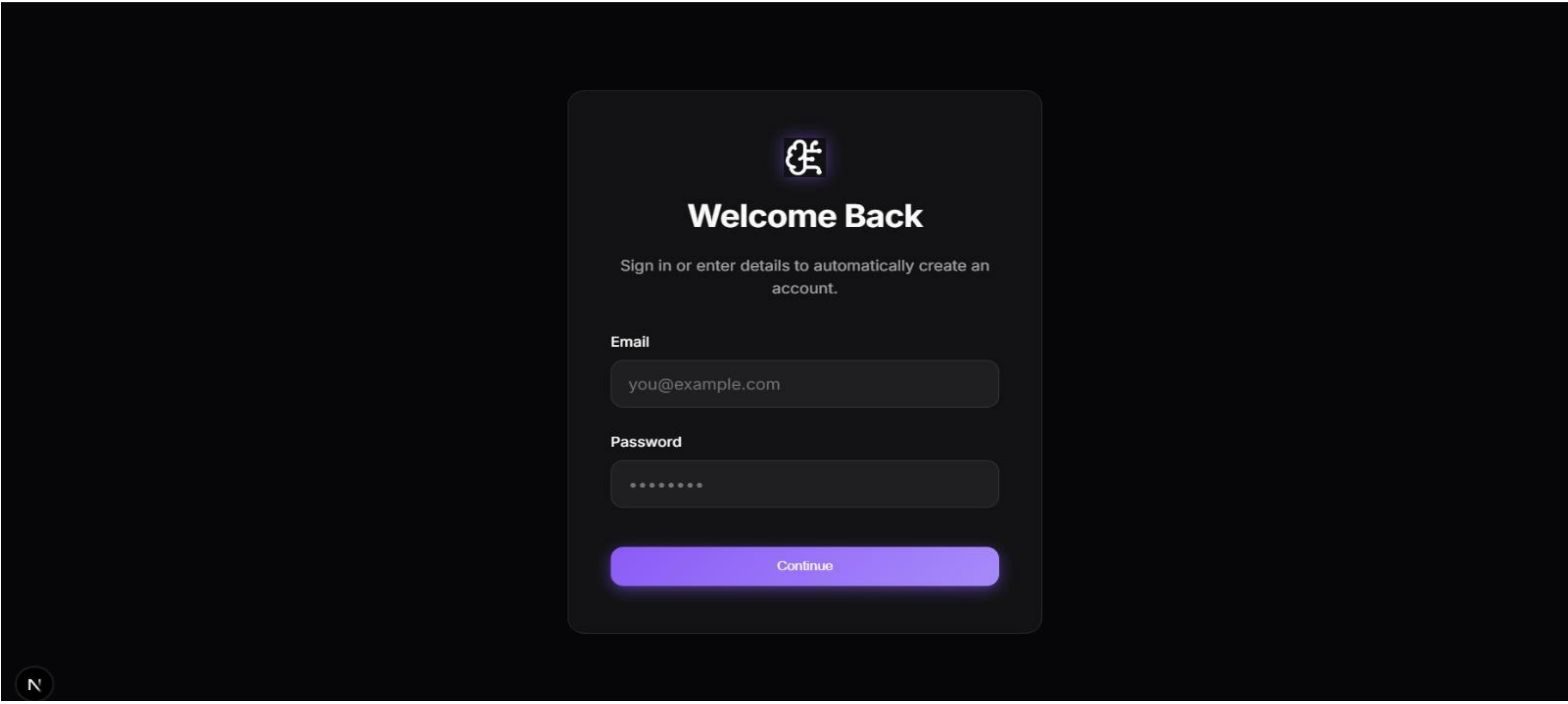
Text Summerization

Experimental Results

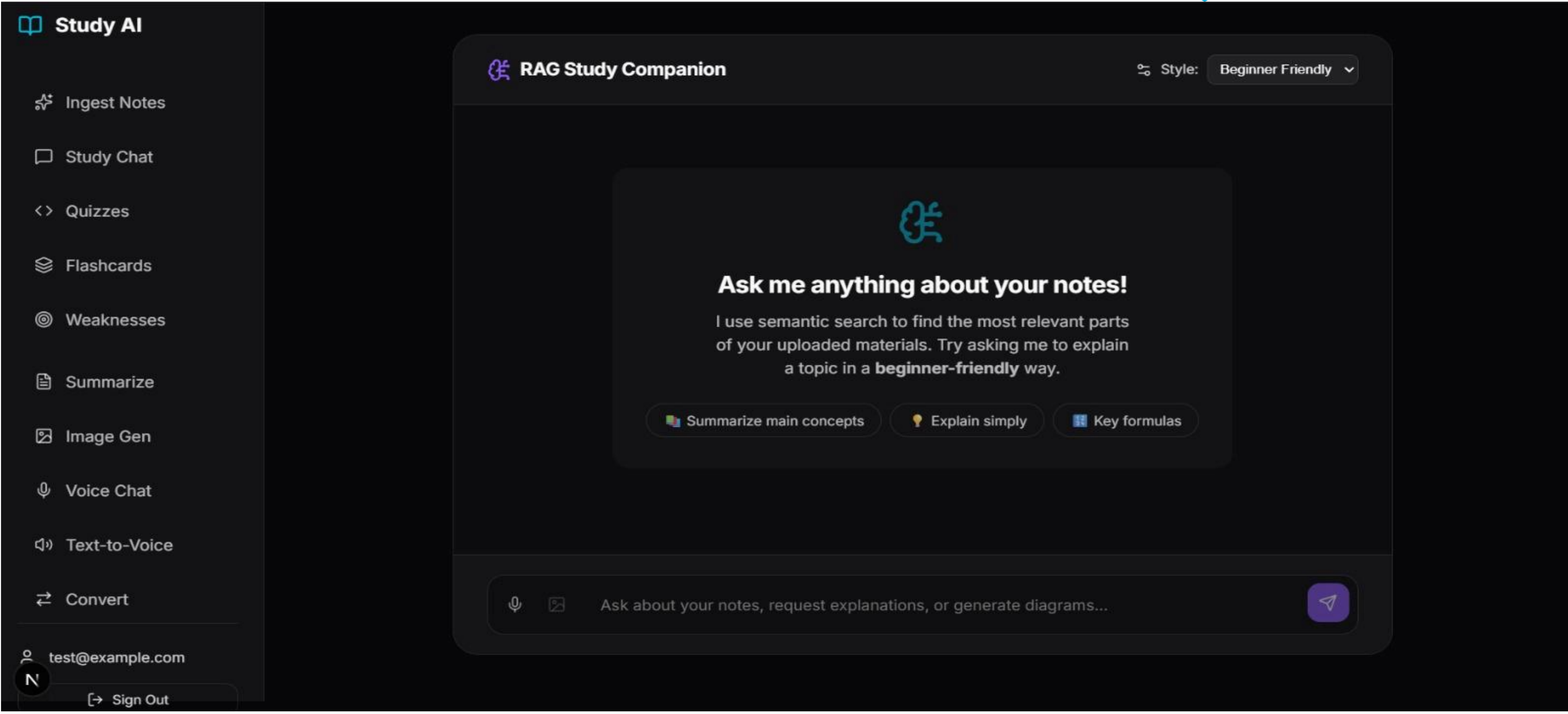
- Faster learning
- Better accuracy
- Reduced manual work



User Login Interface



Rag Interface



Quiz & Viva Generation Interface

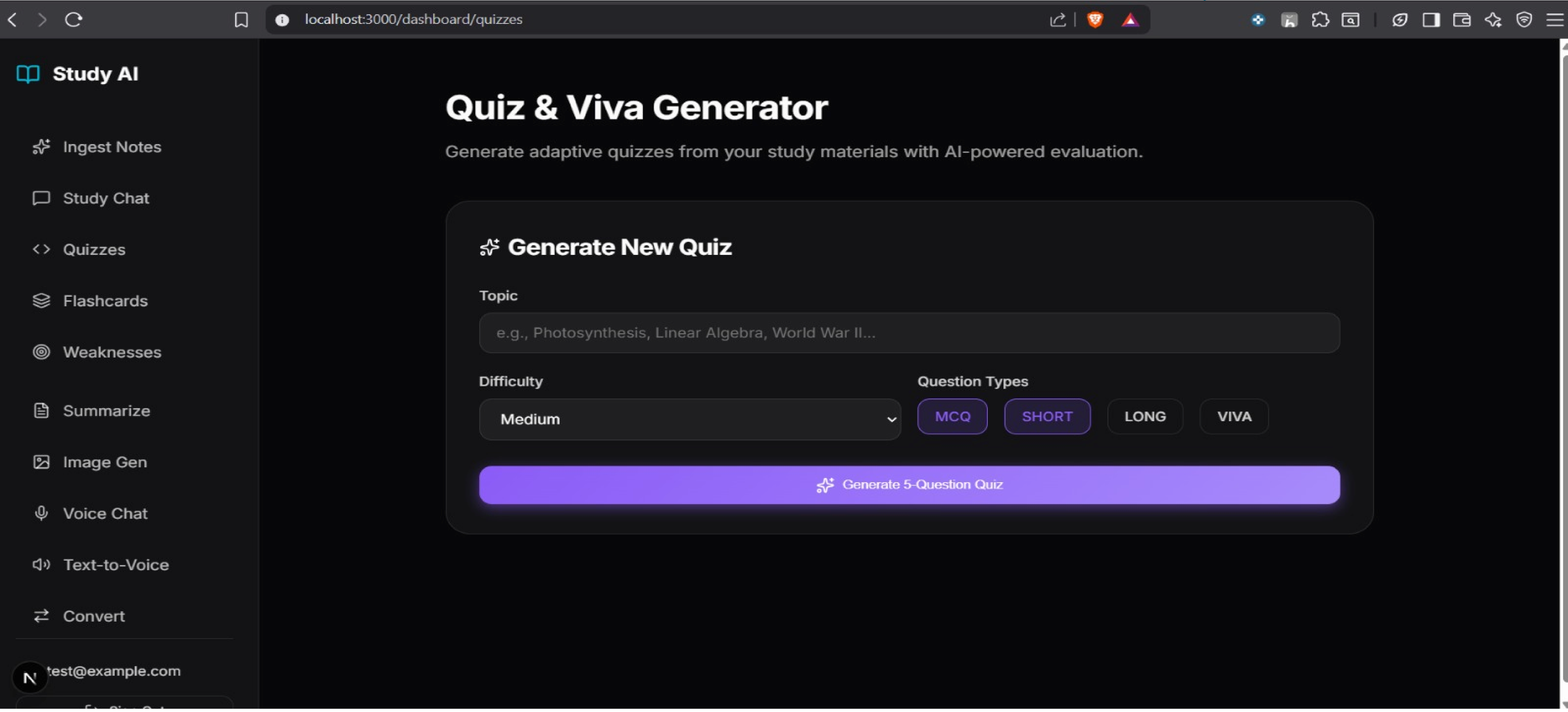
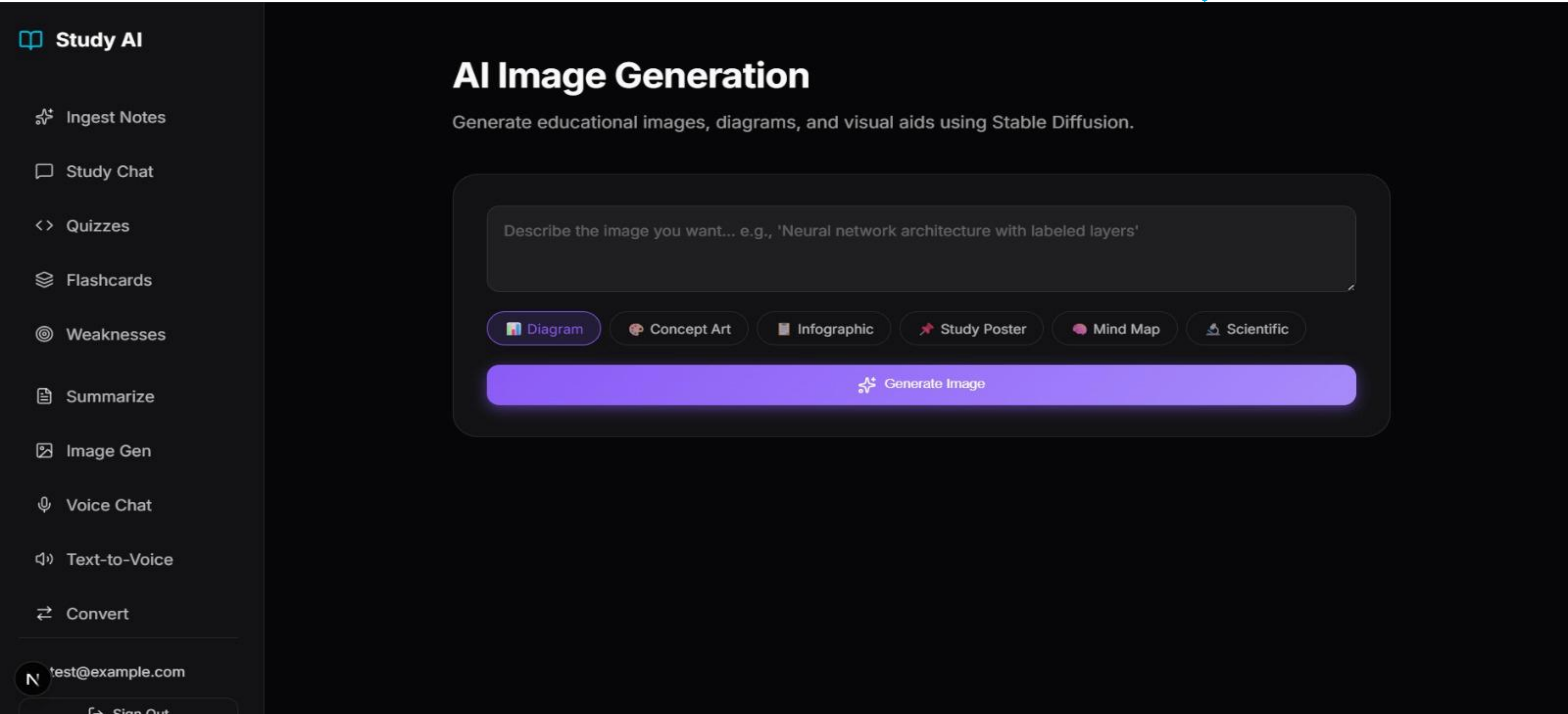
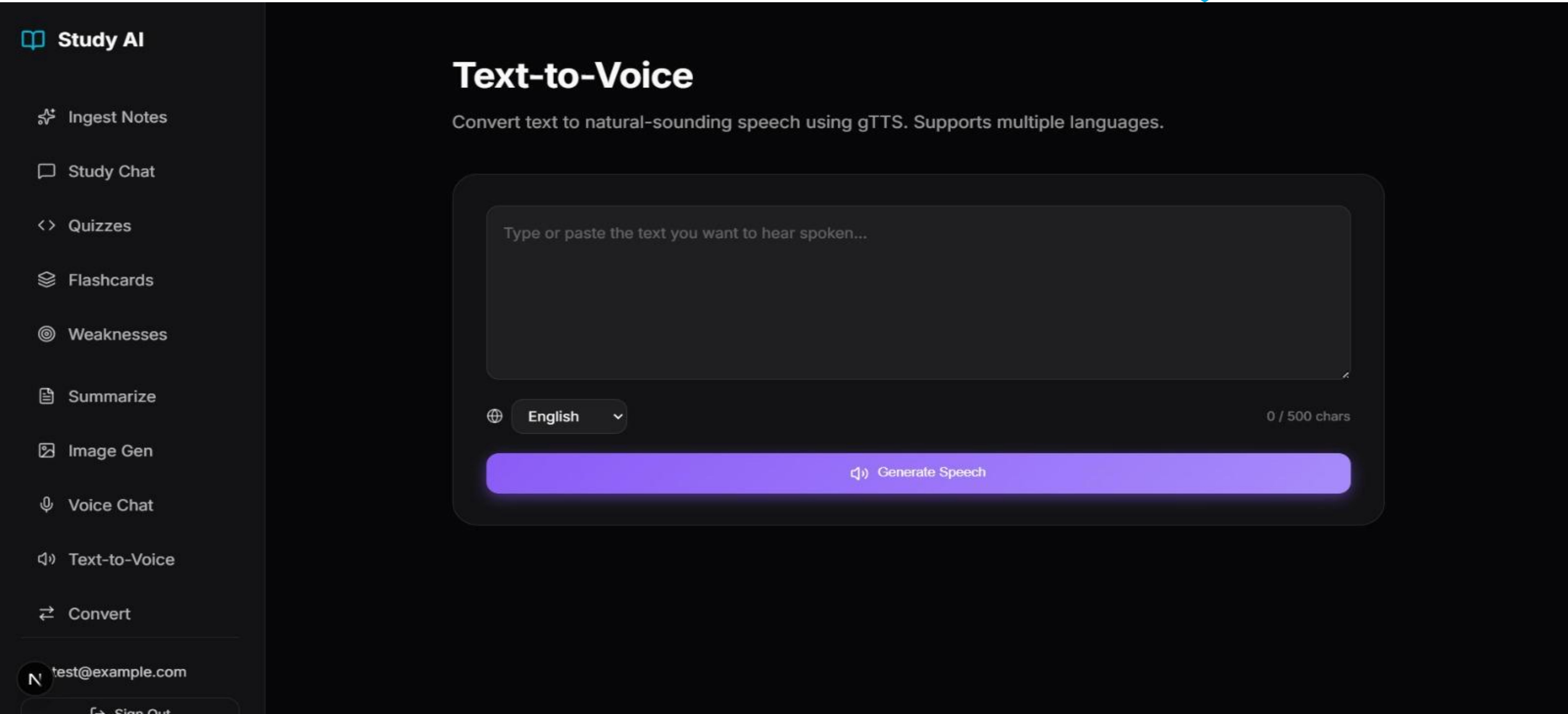


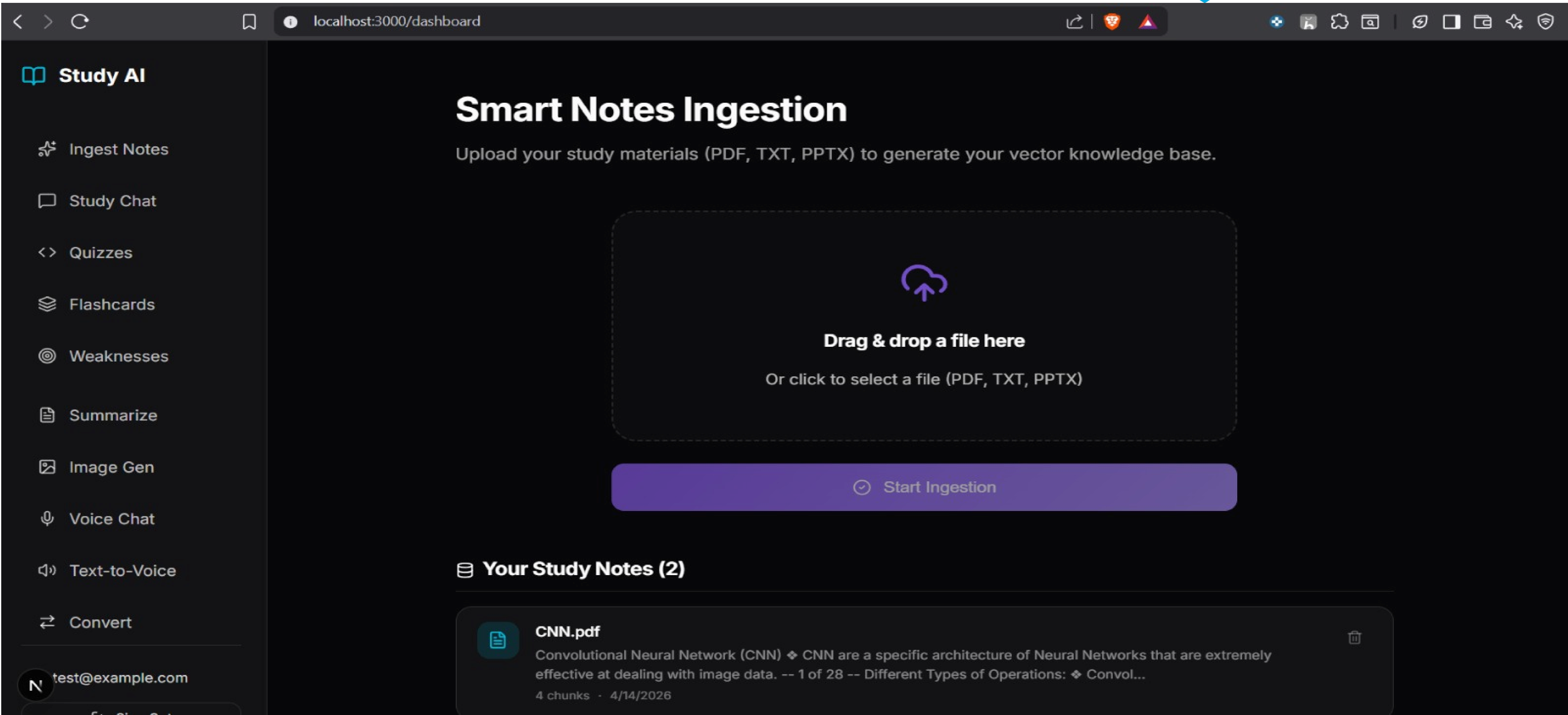
Image Generation By User



Text To Voice Generation Interface



Question Via Notes



Conclusion

•Developed a **multimodal AI-powered study companion**

•Successfully integrates:

•RAG-based learning

•Voice interaction

•Progress tracking

•Addresses limitations of existing systems

•Provides a **complete personalized learning solution**